Ceramics & Composites

Ceramics

Ceramics are inorganic, non-metallic compounds, consisting of metallic and non-metallic elements joined together. With metallic and non-metallic elements, ionic bonds form, while with only non-metallic elements, covalent bonds form.

There are two main groups of ceramics, traditional and engineering ceramics.

Traditional Ceramics

Originally, ceramics were made with simple substances like clay, silica and water, usually used for figurines, pottery, art pieces, and more practical applications like bricks and tiles.

Engineering Ceramics

These ceramics are usually made with pure compounds like aluminium oxide, silicone carbide, and others, depending on their uses, which can be dental implants, bearings gas turbines, and others.

Structure of Ceramics

Generally, the atomic structure of ceramics consists of fully/partially ionic bonds. Ionic bonds are formed by the strong electrostatic attraction between cations and anions. Generally, these stronger ionic bonds cause higher melting points, and harder materials.

For the crystal size of ceramics, they are influenced by the magnitude of the electrical charges, where a larger magnitude of charges causes stronger bonds, and the relative sizes of the ions.

Properties of Ceramics

Generally, ceramics are

hard & brittle

have low toughness/shock resistance

have high compressive strength but low tensile strength

low electrical and thermal conductivity

high melting temperatures, with good chemical & thermal stability

high corrosion resistance

As ions in the ceramic form closely packed crystals, the atomic structure of crystalline ceramics is highly ordered, making it difficult for dislocations to move throughout the material, hence higher stresses are needed to deform or fracture the ceramic.

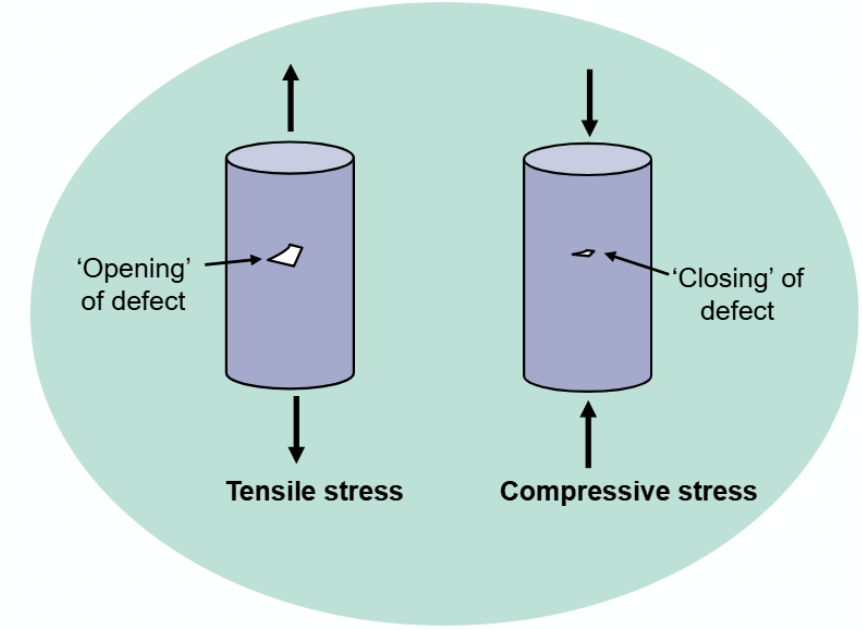
Processes like sintering, or firing, also form stronger bonds in the structure.

The strong bonds in the ceramic cause the high hardness, its high melting point, and chemical stability (also relates to corrosion resistance)

Cracks, pores and inclusions in ceramics often act as stress concentration points. Hence, under loads, ceramics have a high tendency to fracture rather than plastically deform. This is why ceramics are very brittle, have low toughness/shock resistance.

As said earlier, ceramics respond differently to tensile and compressive loads.

Under tensile stresses, defects like inclusions are easily “opened up” due to the tensile load, even under low stresses. Under compressive stresses, defects are closed up, preventing stress concentrations, and allowing stable crack propagation.



Typically, these defects are prevented by controlling the manufacturing process, to optimise its strength and control its brittleness.

Processing of Ceramics

Typically, most metal processing methods involve casting or plastic deformation techniques. However, these are not applicable or practical for ceramics, due to ceramics' high melting point, and brittleness.

Instead, most ceramics are formed by compacting powders (powder production) or particles into shapes (shaping), then heated to high temperatures to bond the particles together (sintering).

Powder Processing

Powder processing involves obtaining high purity powders which are fine and uniform in size, which enhance the shaping process as voids between the particles are minimised.

Additives like binders may be incorporated to help in shape formation, enhancing the mechanical strength of the green ceramics (referring to compacted ceramic powders before firing), and other properties.

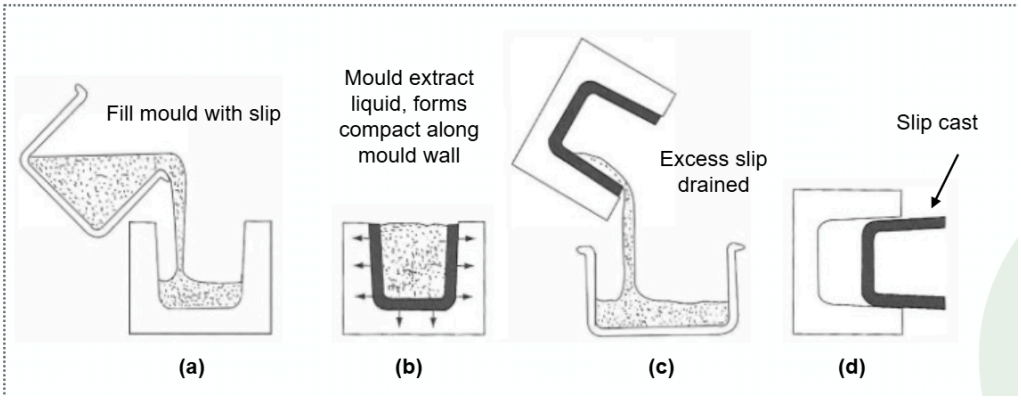
Usually, these powders are made via mechanical methods, being roll crushing, ball milling and hammer milling. Afterwards, screening and sizing is done afterwards to ensure the fine and uniform powder. Powders of advanced engineering ceramics may instead be obtained via chemical reactions, like reductions of their carbides and nitrides.

Shaping

The ceramic powder is then shaped, by compaction or pressing in either a dry or a wet condition, to form green ceramics via processes like slip casting or pressing.

Slip Casting

For slip casting, it is mainly used for producing complex shapes. A slurry of the ceramic powder, known as slip, is poured into a mould, which is able to extract liquid from the slurry in contact with the mould. This forms a compact along the mould wall, allowing the excess slip to be drained, and hence forming a cast, which can be removed after leaving to dry.



Uniaxial Pressing

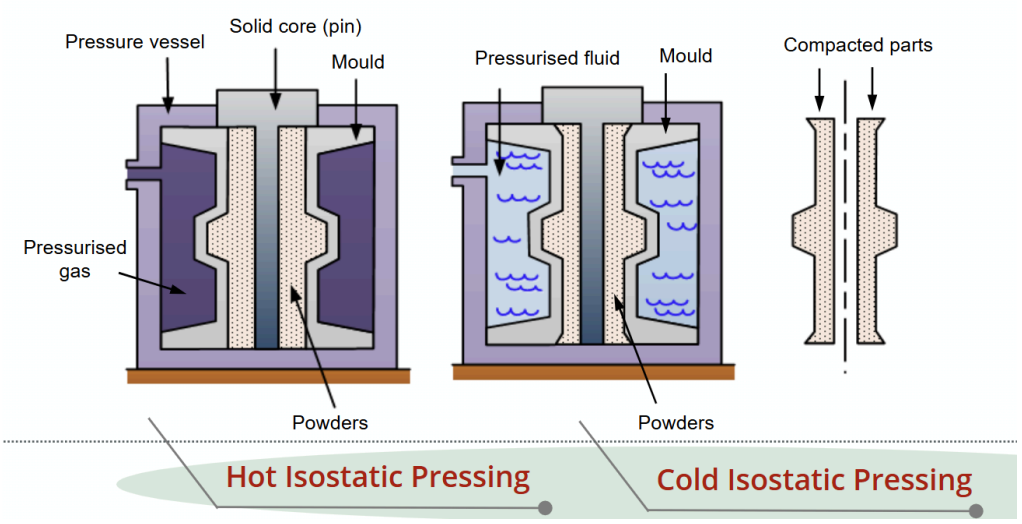
Uniaxial pressing involves pressing the powders along one direction to compress and shape the powders. Powder is placed in a die, and is compressed along one direction by two rigid punches, packing the powders closely. This is mostly used in mass production of simple parts, or for producing advanced ceramics.

Isostatic Pressing

Isostatic refers to pressure being applied uniformly in all directions, either via hot or cold pressing (based on temperature applied).

Hot isostatic pressure is used to enhance the density and mechanical properties of the ceramic, by placing the ceramic powder is a sealed vessel to be subjected to high uniform pressures from an inert gas, like argon, and high temperatures up to 2000C. This causes the powders to be compacted into a dense green body.

Similarly, cold isostatic pressing is placed in a sealed flexible mould to be compressed under uniform pressures via pressurised fluids, like water or oil, at room temperature.

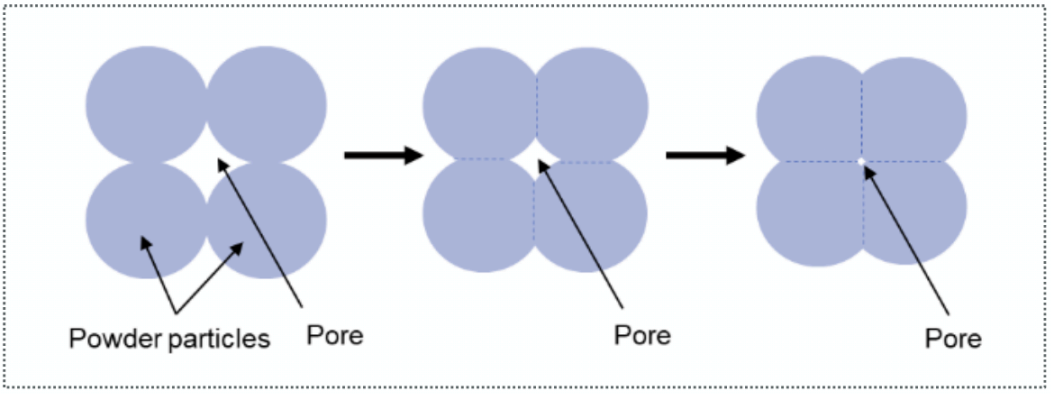


Both isostatic pressing methods help to reduce porosity, providing a uniform dense green body. Having a uniform density also helps to improve mechanical properties like strength and toughness. Isostatic pressing is also better for intricately shaped parts.

Sintering

After shaping, sintering is done to improve the mechanical properties of the final ceramic, by heating the compacted powders to high temperatures but still below the melting point.

By sintering, particles are bonded together, reducing porosity and increasing rigidity and strength



At first, particles are discrete, and contain pores prior to sintering, but as the temperature increases and sintering occurs, the particles diffuse and bond to one another, causing pores to reduce and be closed up. As such, the part will shrink slightly due to the pores closing up.

As sintering continues, densification continues.

Once sintered, the part is then cooled to room temp slowly to avoid thermal shock & cracking

Applications of Ceramics

Applications	Requirements	Examples
Wear parts Seals, bearings, valves, nozzles, cutting tools, abrasive	High hardness High wear resistance High thermal stability Low friction	Alumina Silicon carbide Silicon nitride Zirconia
Heat engines Diesel engines, spark plugs, glow plugs, gas turbine engines	High strength High thermal stability High creep resistance	Alumina Silicon carbide Silicon nitride Zirconia
Biomedical Medical implants	Biocompatibility Durability High wear resistance	Alumina Zirconia Hydroxyapatite Bioactive glass
Teeth replacement	Biocompatibility Durability Aesthetic qualities	Alumina Zirconia Porcelain
Construction Highways, bridges and buildings	Durability Aesthetic qualities	Concrete Advanced cement eg. high performance concrete, fibre-reinforced concrete
Defence Armour materials	High strength Low weight Resistance to extreme condition such as high velocity impact	Alumina Boron carbide Silicon carbide

Composites

Composites are materials that are mixtures of two or more materials to achieve properties that are superior to the properties of their individual materials.

Generally, composites are composed of reinforcements embedded inside a matrix.

A matrix may be a polymer, metal or ceramic, acting as a binder to hold the reinforcements.

Reinforcements can come in the form of particulates, laminates or fibers, and they generally help to provide strength and stiffness to the material.

Overall, the combination of the two provides an increased strength and durability.

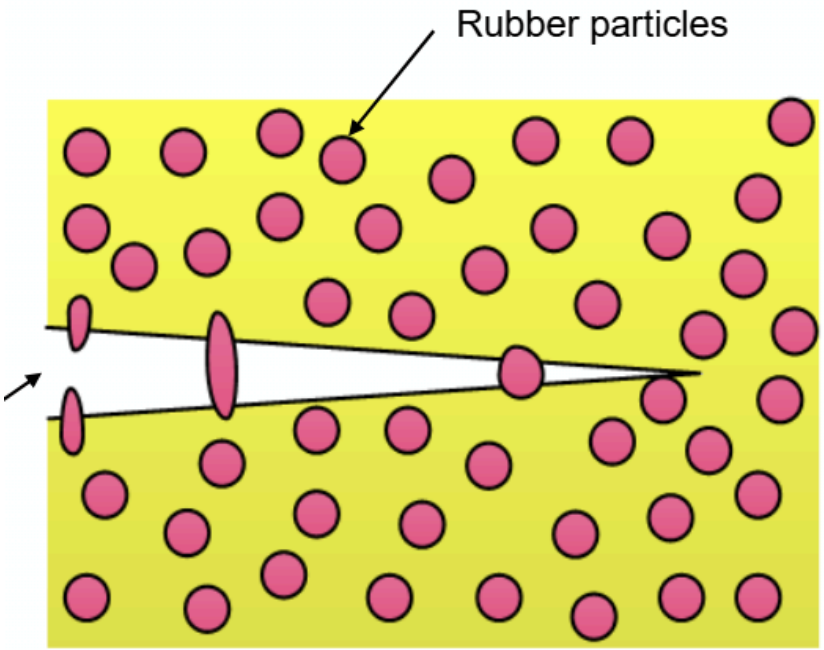
However, factors that affect the properties of the composite are

- the properties of the individual components, [PROPERTIES]
- the size, shape, distribution, and volume fraction of the components, [PROPORTION]
- the bond nature and strength of the bonds between components. [BOND]

As such, composites are grouped by the type of reinforcements they have:

Particulate Composites

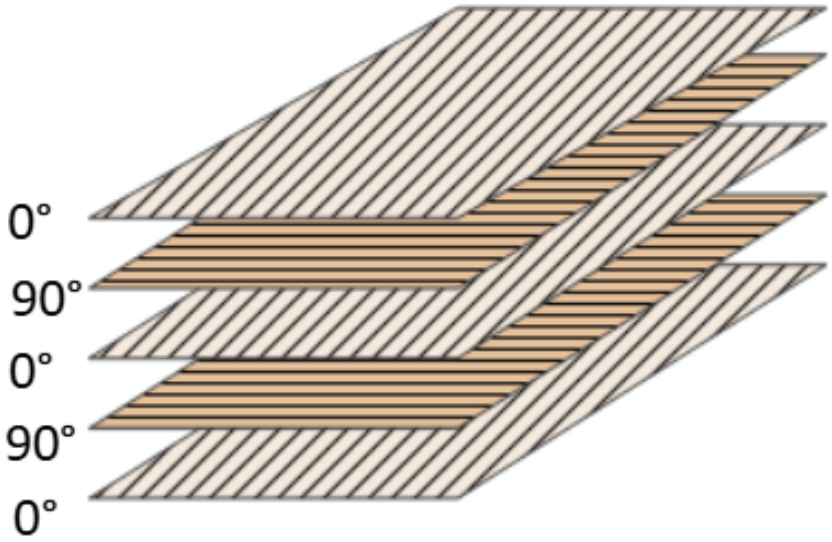
- These consist of small particles, which have improved mechanical properties, dispersed in a comparatively softer and tougher matrix.
- The harder particles help to increase the strength, stiffness and hardness of the composite, as the particles restrict movement and propagation of cracks in the matrix due to the stronger particles.
- However, toughness usually drops, due to the harder but less tough particles. This can be mitigated via using rubber particles. As cracks propagate, the cracks stretch rubber particles, which act as springs, causing the cracks to be clamped shut. Hence, to crack the material, more load is needed to crack material and overcome the clamping force.



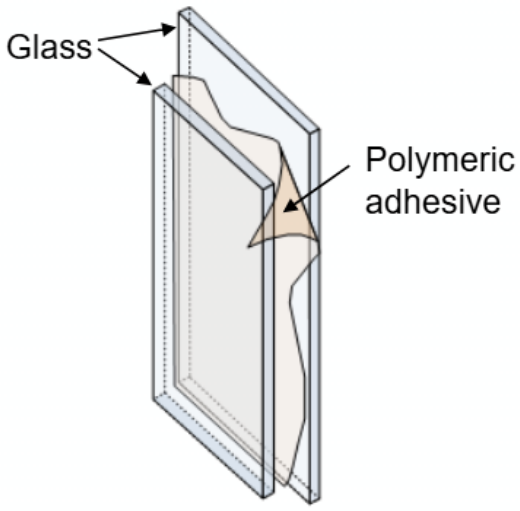
Laminate Composites

Laminate composites consists of layers of materials joined together, and aligned in a specific manner to bear the applied loads effectively, and for other properties.

For example, wood on its own has fibers aligned in only one direction, giving it unidirectional strength and properties. This means that it is only able to withstand loads in one direction, but by laminating the wood at perpendicular directions, it is able to withstand loads in either of the directions, creating high strength in either direction.

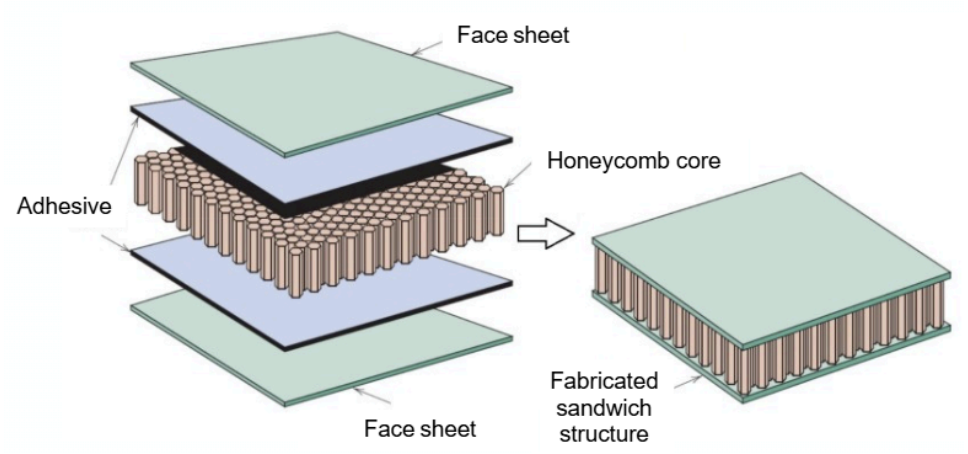


Another example is laminated safety glass, used in car windscreens. They comprise of two glass sheets and a polymer adhesive in between, holding the glass together, thus improving its strength. During fracture, shattering is reduced as fragments are still stuck to the adhesive, preventing fragments from flying out.



Other examples include honeycomb laminate, where a hexagonal cell structure acts as a core for structural support, while sandwiched between two laminate skins, held together by adhesives in between.

This gives it a high strength to weight ratio, making it strong and rigid, while being light, making it useful for aerospace applications.



Fibre-Reinforced Composites

Fibre-Reinforced Composites are comprised of stiff but brittle fibers in a soft but ductile matrix.

The soft matrix helps to bind the fibers together, while also preventing the fiber surface to be damaged or chemically corroded.

The tougher matrix also transmit loads to the fibers, and prevents brittle cracks from spreading throughout.

The transmitted loads are carried by the fibers, which help to improve the strength, stiffness, fatigue resistance, and strength to weight ratio of the composite.

Generally, there are two ways to arrange the fibers in the composite:

Continuous Fibers

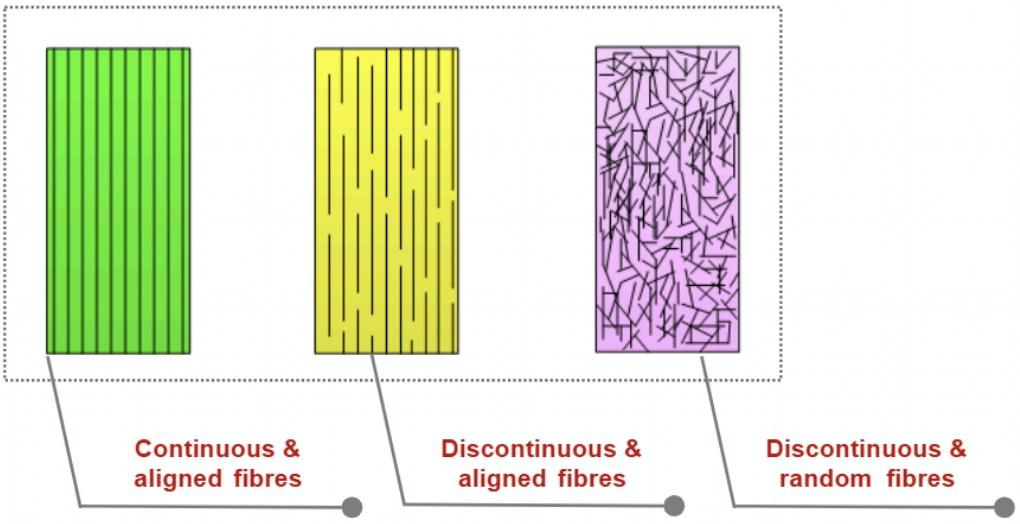
These fibers tend to be longer, that are usually aligned in one direction in the matrix

By aligning the fibers, it allows a maximum strength along the aligned direction, as the fibers can effectively carry loads when applied in the same direction as the fibers.

However, when loads are applied in the transverse direction, the strength can be low, and will not withstand as much load.

Discontinuous Fibers

These fibers tend to be shorter, and may be aligned or randomly oriented. Usually, a random orientation is preferred if the stress direction is unknown, as the randomly oriented fibers help to support loads in all direction, providing even strength in all directions.



Thus, the strength of a fiber reinforced composite is dependent on how the fibers are aligned with the applied stress.

Rule of Mixture

To determine the strength, σ, and stiffnes, E, of the material, the rule of mixture can be used for a composite with n materials:

$$\sigma = \sum_{i=1}^{n} V_i \sigma_i, \;\; E = \sum_{i=1}^{n} V_i E_i$$

where Vi is the volume fraction of each material.

Hence, $0 < V_i < 1$. For the rule of mixture for the area fraction, it is

$$\sigma = \sum_{i=1}^{n} A_i \sigma_i, \;\; E = \sum_{i=1}^{n} A_i E_i$$

Again, Ai is the area fraction of each material, and ranges from 0 to 1.

Generally, there are 3 types of materials for fiber reinforcement.

Glass

Glass fiber reinforced composites, or GFRP, usually consists of a thermoset polymer matrix, like epoxy, polyester resins to bind the glass fibers together, and to embed the glass fibers into the resin matrix

This is the most widely used type of fiber composites, due to the cheap glass fibers, the high availability and ease of machining.

Typically, these are used in non critical performance applications, like pipes, panel roofs, and boat hulls.

Carbon Fiber

Carbon fiber reinforced polymers, or CFRPs, use a thermoset polymer, like polyester resins, to bond carbon fibers together.

CFRPs have a higher strength to weight ratio and stiffness to weight ratio compared to GFRPs, making them useful in the aerospace & sports industries.

Aramid (e.g. Kevlar)

Aramid fiber reinforced polymers, AFRPs, use aramid fibers embedded in a polymer matrix.

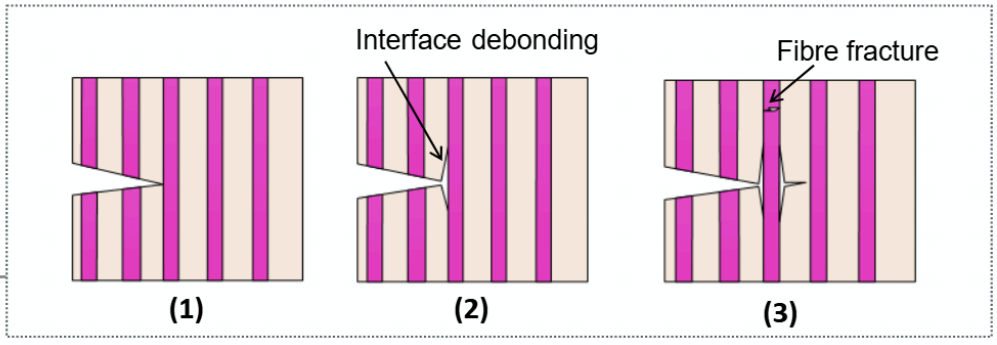
usually, Kevlar is used as the aramid fibers, due to its low density, high impact, heat and abrasion resistance. However, it is the most expensive out of the three fibers.

Its low density makes it suitable for aircraft components, and its impact and abrasion resistance makes it useful for body armour, helmets and handphone covers.

For these fiber reinforced composites, there are two modes where they may fail.

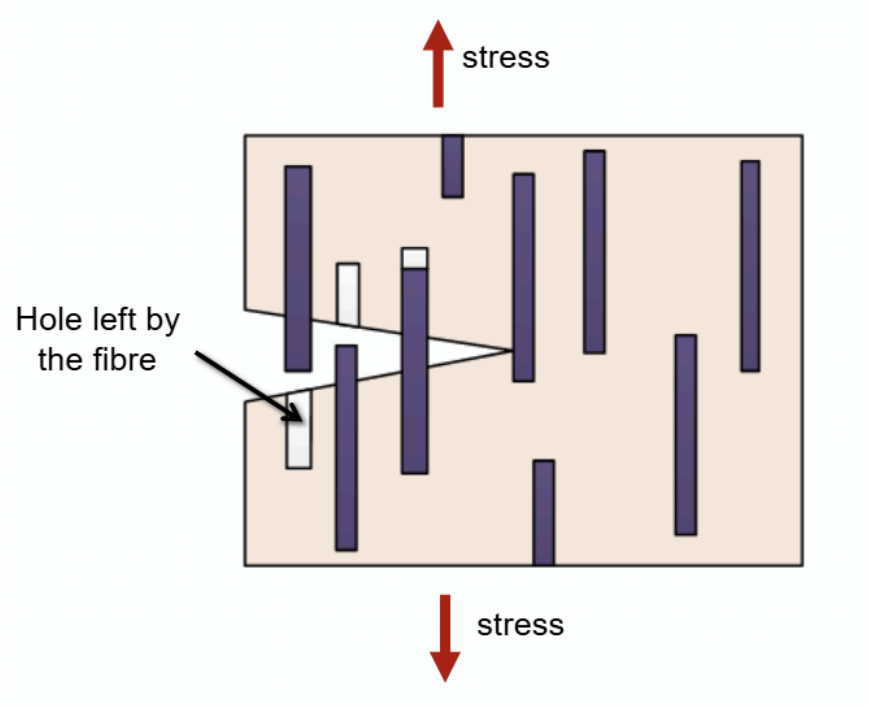
Fiber Fracture

As cracks propagate through matrix, they are deflected at the matrix-fiber interface, causing debonding between the matrix and fiber, causing the fibers to fracture. This prevents the load from being transmitted to the fibers, causing the soft matrix to fail.



Fiber Pullout

When there is poor adhesion between the fibers and matrix, under tensile stress, the fibres are pulled out of matrix, causing lower strength, hence causing the composite fails.



Composites are processed in five common methods:

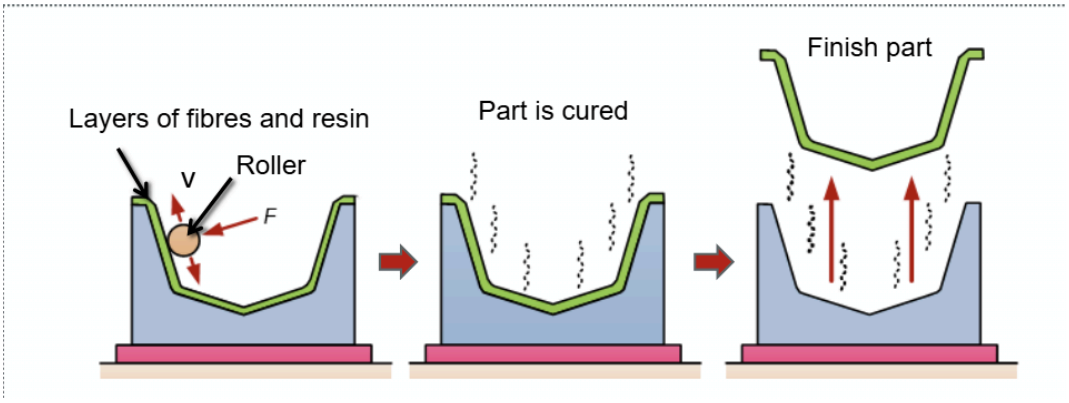
Hand Lay-up

Hand lay-up is a manual method for manufacutring composites. First, a mould is cleaned and treated with a release agent, and the reinforcements used are cut to the appropriate size.

Resin is then applied by rollers or brushes in layers, along with the fibers. Rollers are then used to press the fibers and resins together for adhesion and removing excess air.

This process is then repeated for forming multiple layers, if necessary.

Afterwards, the part is left to cure either at room temperature or in an oven and then removed and trimmed.



This process is most suitable for boat hulls, canoes, tanks, and building panels.

Spray-up

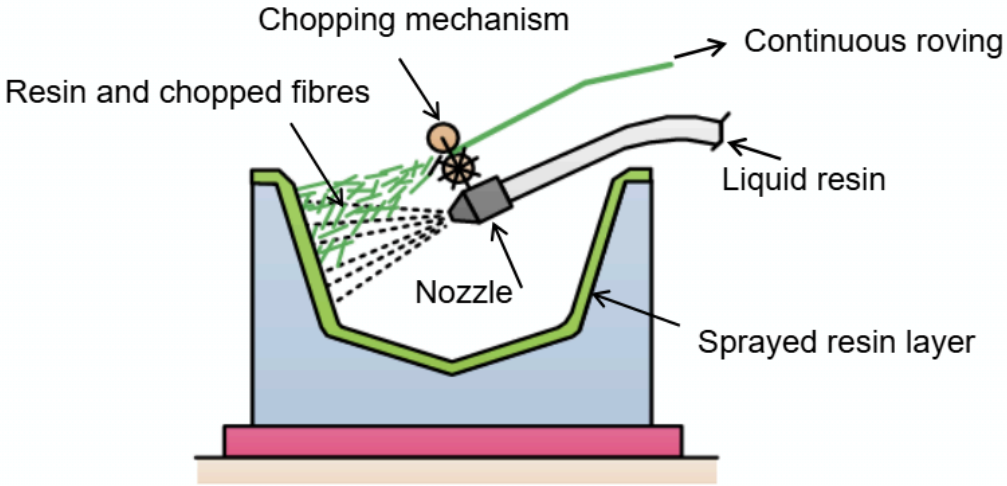
Spray-up is an extension of the hand lay-up method.

Again, the mould is cleaned and treated with a release agent. However, a gel coat is first applied on the mould and left to cure first.

Then, a spray gun applies the resins and a chopper gun chops and applied the fibers simultaneously.

Once a layer has been formed, it is then rolled similarly and further layers may be added afterwards.

The part is then left to cure in similar conditions before being removed.



This process is mainly used for larger parts, like boat hulls, storage tanks, bathtubs, shower units, and custom car parts.

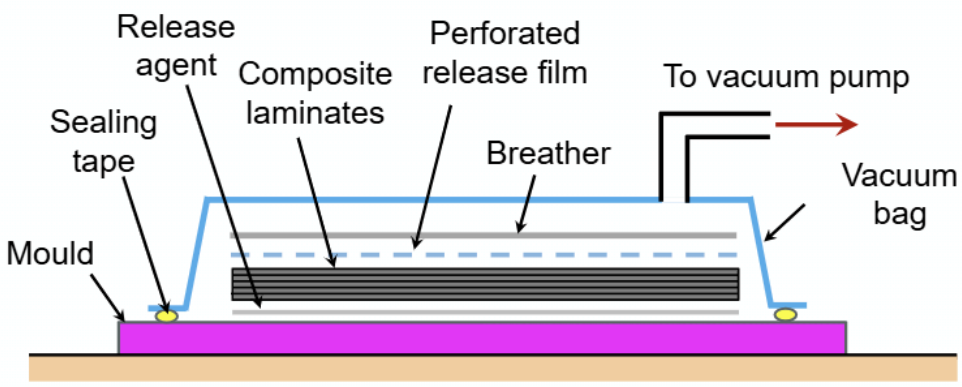
Vacuum Bagging

Vacuum bagging is a manufacturing process meant to improve the quality & strength of composites.

First, reinforcements and resins are placed on a cleaned and treated mould.

A perforated release film is then placed on top, followed by a breather, then sealed inside a vacuum bag.

Air is then evacuated, which provides uniform pressure to the laminate, before being cured in the vacuum at room temp or in an oven.



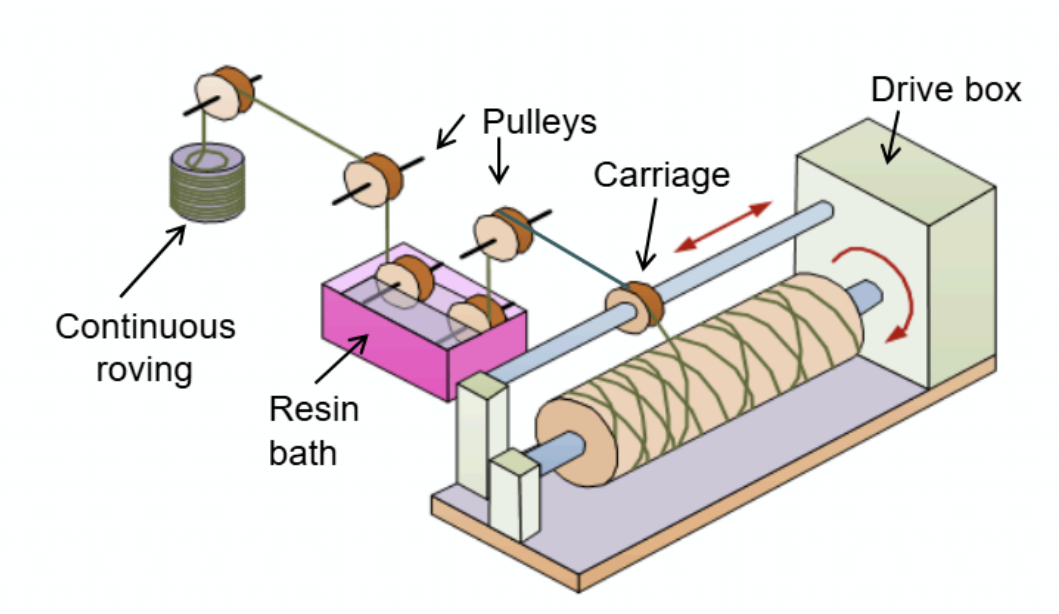
This process gives high strength to weight parts, thus useful for aircraft wings, fuselage, custom car parts, tennis rackets.

Filament Winding

Filament winding is used for manufacturing strong and lightweight composites, typically cylindrical parts. First, a mandrel is cleaned and prepared, and continuous fibers are fed through a resin bath before being woven onto a rotating mandrel. These fibers are laid in specific patterns on the mandrel, for improved strength.

Multiple layers of the resin-impregnated fibers are laid on the mandrel, and their angle and pattern can be adjusted for each of the layers.

Afterwards, the worn fibers are left to cure at room temperature or oven to harden and cure, before being removed from mandrel

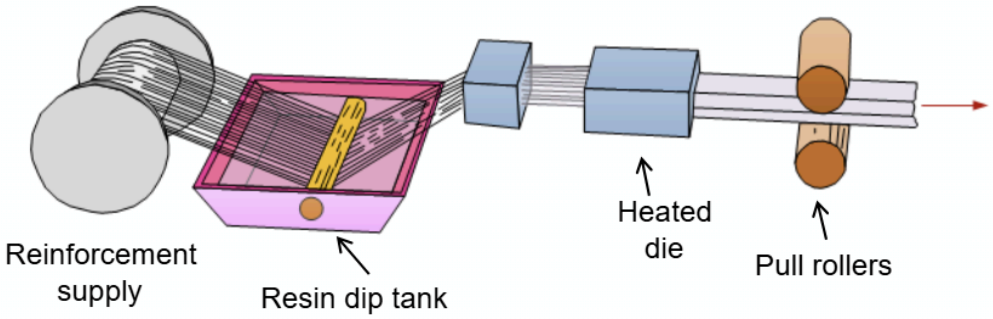


This process is most useful for cylindrical profile parts, like pressure vessels, pressure pipes, filter housings

Pultrusion

Pultrusion is a continuous manufacturing process used for making constant cross section profile parts.

Fibers are pulled from spools and fed through resin deep bath to have them impregnated with resins. The fibers are then guided to a heated die, where the resins cure and harden into the desired profile. Afterwards, the pulling mechanism draws cured profile out, for it to be cut to length.



This process helps to produce efficient, strong and lightweight parts, useful for structural beams, rods and tubes.